Archivara Research Pipeline via Codex CLI

Archivara Agent: A No-Go Checkpoint and Transfer-Safe Obstruction-Atlas Program for $R(5,5)$

March 13, 2026

Claim class. Infrastructure-only synthesis plus a negative-result checkpoint.

Current frontier. $43 \leq R(5,5) \leq 46$.

Supported outcome. A fully specified, falsifiable obstruction-atlas route and a proof that the present repository state cannot justify either bound movement or an achieved structural intermediate result.

Abstract

The current published frontier for the diagonal Ramsey number $R(5,5)$ is $43 \leq R(5,5) \leq 46$ [4, 8, 3, 15]. The central scientific question is not whether one can describe another optimizer or another SAT configuration, but whether one can identify a reusable Ramsey object that links constructive lower-bound work to exact upper-bound proofs. This paper studies that question under the exact evidence available in the local Archivara repository. We synthesize the curated Ramsey corpus, concept-evolution outputs, verification packet, figure packet, commit history, and decision log, and we formalize the only route that remains novelty-safe on the current record: a transfer-safe obstruction atlas extracted from failed $42 \to 43$ extensions. Our main result is a repository-state no-go theorem: because no `results/h1/*` evidence layer exists, no independently verified 44-vertex witness exists, and no machine-checkable 45-vertex impossibility proof exists, the repository supports neither a bound-improvement claim nor an achieved H1 structural result. What it does support is a rigorous research program: a canonical data model, an exact rung-by-rung evaluation ladder, explicit kill switches, and a claim grammar that blocks proxy metrics from being misreported as progress. The significance is methodological but nontrivial: for R(5,5), the repository now contains a falsifiable structural program and a disciplined stop signal instead of an open-ended narrative of apparent progress.

1 Introduction

The diagonal Ramsey number R(5,5) sits at a narrow and stubborn frontier. Exoo's constructive lower-bound work established $R(5,5) \geq 43$ [4]; subsequent exact upper-bound work pushed the best published upper bound from 53 to 48 and then to 46 [13, 2, 3]. The resulting gap,

$$43 \leq R(5,5) \leq 46,$$

remains one of the cleanest places where modern exact computation, constructive search, and proof infrastructure all meet, but none yet dominates.

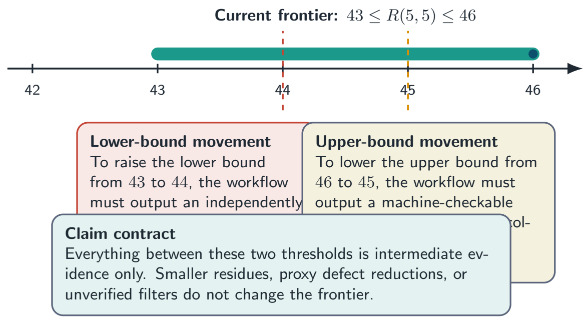
The repository studied here was created for the task "Improve the Ramsey number $R(5,5)$ bound." A naive reading of that prompt would invite a result-first manuscript. The verification packet rules that out. The benchmark layer is empty, the planned H1 artifact layer `results/h1/*` does not exist, and the accepted claim grammar allows no wording that would blur proxy improvements into actual frontier movement. The correct scientific task is therefore narrower and sharper: identify the strongest claim class that is currently defensible, state it formally, and show exactly what evidence would be required to go further.

The repository converges on a single plausible novelty route. The idea is not another one-vertex-extension engine, another low-defect search heuristic, or another proof logger. The proposed object is a *cross-family canonical obstruction atlas* for failed $42 \to 43$ extensions: if many failed extensions share a small dictionary of minimal obstruction cores that transfer across held-out parent families and preserve all known witnesses, then those cores could become a structural bridge between the lower-bound and upper-bound lines. If they do not transfer, the route should be killed quickly.

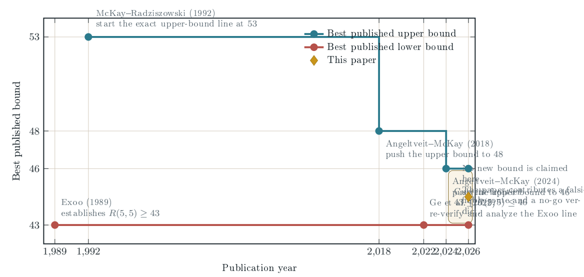
This paper makes four contributions.

1. It reconstructs the strongest repository-supported route as a full research program: a precise data model, exact evaluation ladder, and explicit kill-switch logic for a transfer-safe obstruction atlas over failed $42 \to 43$ extensions.

2. It proves protocol-level results that the repository implicitly depends on but did not state as theorems: quotient canonicalization removes representation noise, witness-oracle safety is the



(a) The claim contract for R(5,5). Anything short of an independently verified 44-vertex witness or a checked 45-vertex impossibility proof remains intermediate evidence.



(b) Historical compression of the published frontier from Exoo's 43-vertex lower bound to the current upper bound 46, with the present manuscript explicitly marked as a no-go checkpoint rather than a new point on the frontier.

Figure 1: Frontier geometry and claim contract. Panel (a) shows the only two events that count as true bound movement in the repository's claim grammar. Panel (b) places the current paper on the timeline of published results: it contributes a falsifiable route specification and a disciplined stop signal, not a new bound. Together the panels explain why an honest R(5,5) manuscript must separate structural route design from bound movement.

correct soundness notion for the current filter bank, and the acceptance contract separates structural intermediate evidence from true bound movement.

3. It gives a publication-grade synthesis of the real prior-art surface, the branch chronology, and the evidence inventory, using only claims that are anchored to the actual artifacts and bibliography.
4. It proves a repository-state no-go theorem: on the present record, no manuscript may honestly claim either a new bound on R(5,5) or an achieved H1 structural intermediate result.

The paper is intentionally explicit about what it does *not* do. It does not improve the lower bound to 44. It does not lower the upper bound to 45. It does not demonstrate a working obstruction atlas. It demonstrates something subtler but necessary: that the route can be specified rigorously, and that the current repository state admits only a constrained no-go / planning paper.

The remainder of the paper is organized as follows. Section 2 locates the obstruction-atlas route among the closest lower-bound, one-vertex-extension, upper-bound, and formal-proof lines. Section 3 defines the objects and claim classes used throughout. Section 4 presents the full proposed pipeline, pseudocode, complexity accounting, and design rationale. Section 5 states and proves the protocol-level results, culminating in the repository-state no-go theorem. Section 6 describes the deterministic repository-measurement setup. Section 7 reports the exact evidence inventory, route triage, and chronology. Section 8 analyzes limitations, failure modes, open questions, and societal implications, and Section 9 concludes.

2 Related Work

The repository's own novelty analysis is correct about one point and easy to misunderstand about another. The correct point is that the closest overlaps are not generic optimization or generic SAT, but a small Ramsey-specific set: Exoo's constructive lower-bound line, Ge et al.'s re-analysis of that line, Lehavi's one-vertex-extension and checking framework, the McKay–Radziszowski to Angeltveit–McKay upper-bound line, Gauthier's recent strategy note on continuing that decomposition surface, and the formal-proof / certificate line around $R(4,5) = 25$ [4, 8, 11, 13, 2, 3, 5, 6, 7, 14]. The easy misunderstanding is to think that the present paper sits on any one of those surfaces directly. It does not. Its positive content is a route target and its negative content is a no-go theorem.

Lower-bound witness construction and analysis. Exoo's 1989 construction remains the best published lower-bound witness family for R(5,5) [4]. Ge et al. revisit that line, re-verify the Exoo witness, and analyze low-defect 43-vertex near-miss structure [8]. Those papers are indispensable baselines, but they do not provide the object targeted here: a cross-family atlas of recurring failure cores across failed $42 \to 43$ extensions. This difference matters. If the present route only redigests Exoo-line structure or only finds new near misses, it collapses back into the Ge/Exoo surface.

One-vertex extension and counterexample checking. Lehavi's 2024 work is the closest direct overlap to any claim about failed $42 \to 43$ extensions because it operates exactly on the one-vertex-extension and counterexample-checking surface [11, 10]. The obstruction-atlas route is novelty-safe only if it produces a reusable structural object beyond extension tables themselves. If it merely catalogs failures or reconstructs the $R(5,5,42)$ to $R(5,5,43)$ transition more cleanly, it is best understood as a repackaging of Lehavi's surface rather than a new contribution.

Upper-bound decomposition. The exact upper-bound line begins with McKay and Radziszowski's case-analysis work [13], continues through Angeltveit and McKay's certified push to $R(5,5) \leq 48$ [2], and reaches the current best published upper bound $R(5,5) \leq 46$ in 2024 [3]. Gauthier's AITP 2025 strategy note indicates that the split-vertex / transverse-edge gluing surface is still the live upper-bound continuation point [5]. This is why the repository demotes H2 to backup status: absent a genuinely different decomposition primitive, H2 risks being "the same proof object with a stronger engine."

Formal proof and certificate infrastructure. The formal-proof line matters because the repository explicitly treats proof packaging as infrastructure unless it attaches to a transferable structural object. The relevant anchors are the formal proof of $R(4,5) = 25$ [6, 7], the Lean 4 formalization of finite Ramsey theory [14], proof-logging practice exemplified by *Schur Number Five* [9], and recent verified SAT+CAS Ramsey certificates [12]. These works show that certificate infrastructure can be first-rate. They also show why the present paper cannot claim novelty from proof packaging alone.

What the repository intentionally avoids. The prior-art gap file and falsifier memo explicitly reject several tempting but weak narratives: generic GA or symmetry stories, generic rare-event narratives, solver-shopping, and pure proof export [1]. The point is not that these directions are valueless in general. It is that, within the current packet, they do not define a novelty moat for R(5,5).

3 Background and Preliminaries

This section defines the mathematical objects and the repository-specific claim classes needed for the proofs.

3.1 Ramsey notation

For $n \in \mathbb{N}$, let K_n denote the complete graph on vertex set $[n] = \{1, 2, \ldots, n\}$. A *two-coloring* of K_n is a map

$$\chi : \binom{[n]}{2} \to \{\text{red}, \text{blue}\}.$$

A coloring is $(5, 5)$-*admissible* if it contains neither a red K_5 nor a blue K_5. By definition,

$$R(5, 5) = \min\{n : \text{every two-coloring of } K_n \text{ contains a monochromatic } K_5\}.$$

The present published frontier is $43 \leq R(5, 5) \leq 46$ [4, 8, 3, 15].

3.2 Objects for the obstruction-atlas route

The planned H1 route is built around failed one-vertex extensions of known 42-vertex frontier graphs.

Definition 3.1 (Frontier parent)**.** *A* frontier parent *is a known* $(5, 5)$*-admissible two-coloring of* K_{42} *together with provenance metadata, a canonical labeling, automorphism information, an orbit partition of its vertices, and a family label indicating which construction lineage produced it.*

Definition 3.2 (Orbit-distinct extension case)**.** *Fix a frontier parent* P*. An* orbit-distinct extension case *of* P *is an equivalence class of ways to adjoin a new vertex* v_{43} *to* P *up to the action of the automorphism group of* P *on the incident red/blue neighborhood pattern of* v_{43}*. One representative from each equivalence class is stored as an extension case.*

Definition 3.3 (Failure witness)**.** *A* failure witness *for an extension case* e *is a minimal set of vertices and incident colors whose presence forces a monochromatic* K_5 *in the extended coloring. The witness is* minimal *if deleting any one of its required incidences destroys the forcing property.*

Definition 3.4 (Canonical obstruction core)**.** *Let* $\mathcal{W}$ *be the set of raw failure witnesses. Define an equivalence relation* $\sim$ *on* $\mathcal{W}$ *by declaring* $w_1 \sim w_2$ *when* w_1 *can be transformed into* w_2 *by a combination of parent automorphisms, relabeling of equivalent extension representatives, and any encoding permutation that preserves the induced forcing pattern. A* canonical obstruction core *is one equivalence class in the quotient* $\mathcal{W}/\sim$ *together with a canonical identifier.*

Definition 3.5 (Family partition)**.** *A* family partition *of the frontier-parent set is a partition*

$$\mathcal{P} = \bigsqcup_{f \in \mathcal{F}} \mathcal{P}_f$$

into non-isomorphic construction lineages. The repository requires at least three non-isomorphic families for any structural H1 *claim.*

3.3 Metrics and claim classes

The repository does not permit generic words like "progress" or "improvement" to float free of evidence. We therefore formalize the relevant metrics.

Definition 3.6 (Coverage)**.** *Let $\mathcal{S}$ be a finite set of failed extension cases, and let C be a set of canonical cores. For each failed extension case $e \in \mathcal{S}$, let $\kappa(e)$ be the set of canonical core identifiers extracted from the witnesses of e. The* coverage *of C on $\mathcal{S}$ is*

$$\mathrm{cov}(C, \mathcal{S}) = \frac{|\{e \in \mathcal{S} : \kappa(e) \cap C \neq \varnothing\}|}{|\mathcal{S}|}.$$

If $\mathcal{S} = \varnothing$, we leave $\mathrm{cov}(C, \mathcal{S})$ undefined because no empirical support exists.

Definition 3.7 (Held-out retention)**.** *Let $\mathcal{F}$ be a family partition, and fix a held-out family $f \in \mathcal{F}$. Suppose a canonical core library C_{-f} is mined from all families except f. The* held-out retention *on f is*

$$\mathrm{ret}_f(C_{-f}) = \frac{\mathrm{cov}(C_{-f}, \mathcal{S}_f)}{\mathrm{cov}(C_{-f}, \mathcal{S}_{-f})},$$

provided the denominator is positive. Here $\mathcal{S}_f$ is the failed-extension set for family f and $\mathcal{S}_{-f}$ is the failed-extension set for the training families.

Definition 3.8 (Witness oracle and witness-safety)**.** *Let Ω denote the stored set of known 42- and 43-vertex witnesses that must survive every proposed filter. A core-derived filter bank B is* witness-safe relative to Ω *if no witness in Ω is rejected by B.*

Definition 3.9 (Claim classes)**.** *The repository admits exactly four claim classes.*

1. *Bound improvement: either an independently verified 44-vertex witness or a machine-checkable proof that all 45-vertex colorings contain a monochromatic K_5.*
2. *Structural intermediate result: a transferable obstruction atlas, decomposition primitive, or certificate object whose transfer and soundness gates have passed.*
3. *Negative result: an explicit route failure under a named stop rule.*
4. *Infrastructure-only result: corpus reconstruction, verification packaging, figure generation, or claim governance without bound movement.*

Definition 3.10 (H1 acceptance contract)**.** *The planned* H1 *route qualifies as a structural intermediate result only if all of the following hold simultaneously:*

1. *the frontier-parent, extension-case, failure-witness, obstruction-core, transfer-record, atlas-summary, and witness-safety artifacts all exist;*
2. *the top recurring cores cover at least 30% of failed orbit-distinct $42 \to 43$ extensions across at least three non-isomorphic families;*
3. *on the decisive leave-one-parent-out anti-Exoo holdout, held-out retention is at least 50% of the training-side coverage;*
4. *zero known witnesses are deleted; and*
5. *no single family contributes more than 50% of the accepted recurring-core support.*

The purpose of these definitions is not literary. They allow the next section to state the protocol-level results that make the current paper mathematically precise.

4 Method

The repository's method is a planned but unexecuted pipeline. That distinction matters: in this section we describe the complete algorithmic program, not an implemented experiment.

4.1 Input corpus and exclusions

The fixed Ramsey-specific corpus consists of the dynamic survey for the current frontier, Exoo's lower-bound paper, Ge et al.'s modern study of the Exoo line, Lehavi's one-vertex-extension and checking paper, the Angeltveit–McKay upper-bound line, McKay–Radziszowski's upper-bound ancestor, Gauthier's strategy note, and the formal-proof / certificate literature around $R(4,5) = 25$ [4, 8, 11, 13, 2, 3, 5, 6, 7, 14, 9, 12]. The repository explicitly excludes the lexical watchlist false positives that arose from a broad seed query. Those exclusions are part of the method because novelty judgments are invalid if the wrong corpus is allowed to carry argumentative weight.

4.2 Pipeline overview

The high-level pipeline is shown in figure 3. It has five stages:

1. reconstruct a local corpus of frontier parents from the Exoo / Ge / Lehavi line;
2. enumerate orbit-distinct one-vertex extension cases from those parents;
3. extract minimal failure witnesses from rejected extensions;
4. quotient those witnesses into canonical obstruction cores; and
5. evaluate held-out transfer and witness safety before allowing any stronger claim.

Two design features dominate the repository's reasoning. First, the pipeline is *transfer-first*: the decisive falsifier is anti-Exoo leave-one-parent-out retention. Second, the pipeline is *soundness-first*: a filter that deletes even one known witness is dead regardless of any apparent coverage gain.

4.3 Algorithmic specification

Algorithm 1 specifies the frontier-reconstruction stage. The key point is that every field written by the algorithm has a named destination in the planned artifact layer. That is why the pipeline is auditable even before execution.

Algorithm 2 describes the core-mining stage.

Algorithm 3 encodes the evaluation logic. It is this algorithm, not a local optimizer, that separates a structural object from a family-specific digest.

4.4 Complexity accounting

Let $p = |\mathcal{P}|$ be the number of frontier parents, let e_i be the number of orbit-distinct one-vertex extensions of parent P_i, and let w be the total number of raw failure witnesses extracted across all failed extensions. Write $T_{\text{can}}(P_i)$ for the cost of canonicalizing parent P_i, $T_{\text{ext}}(P_i)$ for the amortized

Algorithm 1 Frontier reconstruction for the planned H1 atlas

Require: Fixed corpus $\mathcal{L} = \{\text{Exoo 1989, Ge 2022, Lehavi 2024}\}$
Ensure: Parent set $\mathcal{P}$, extension-case set $\mathcal{E}$, witness multiset $\mathcal{W}$

```
P ← ∅, E ← ∅, W ← ∅
for each source s ∈ L do
    import or reconstruct every frontier parent described by s
    for each imported parent P do
        compute a canonical label, automorphism group summary, degree profile, and family label
        append the resulting record to P
        enumerate one-vertex extensions of P up to orbit equivalence
        for each orbit-distinct extension case e do
            append the extension metadata to E
            if e is not (5,5)-admissible then
                extract one or more minimal monochromatic-K_5 forcing witnesses for e
                append those witnesses to W
            end if
        end for
    end for
end for
return (P, E, W)
```

cost of generating an orbit-distinct extension representative of P_i, and $T_{\text{wit}}(e)$ for the cost of shrinking one failed extension e to minimal witnesses.

Then Algorithm 1 has total cost

$$\sum_{i=1}^{p} \left(T_{\text{can}}(P_i) + e_i \, T_{\text{ext}}(P_i) + \sum_{e \in E(P_i)} T_{\text{wit}}(e) \right).$$

Algorithm 2 runs in

$$O\left(\sum_{w' \in \mathcal{W}} (T_{\min}(w') + T_{\kappa}(w')) \right),$$

where $T_{\min}$ is the cost of witness minimization and T_{κ} the cost of canonical hashing. Algorithm 3 adds

$$O(|\mathcal{F}| \, T_{\text{atlas}} + |\Omega| \, |B| \, T_{\text{embed}}),$$

where T_{atlas} is the cost of rebuilding the atlas on a training split, $|B|$ is the number of candidate filters, and T_{embed} is the cost of checking whether a witness embeds any filter-triggering core.

These formulas are intentionally symbolic. The repository does not contain the missing enumerator, witness verifier, or certificate checker needed to instantiate them numerically.

4.5 Design rationale

Every non-obvious design choice in the repository exists to kill a false positive quickly.

- **Canonicalization first.** Without quotienting by automorphism and encoding equivalence, raw recurrence is uninterpretable because it may be inflated by representation noise.

Algorithm 2 Canonical obstruction-atlas construction

Require: Witness multiset $\mathcal{W}$ and equivalence relation $\sim$
Ensure: Canonical core library C with support statistics
  $C \leftarrow \varnothing$
  for each raw witness $w \in \mathcal{W}$ **do**
    shrink w to a minimal forcing witness $w_{\min}$
    compute its canonical identifier $\kappa(w_{\min}) \in \mathcal{W}/\sim$
    increment the support count and provenance counts of $\kappa(w_{\min})$
  end for
  sort canonical cores by support
  return the resulting dictionary together with per-family support totals and raw-versus-canonical compression statistics

Algorithm 3 Held-out transfer and witness-safety audit

Require: Family partition $\mathcal{F}$, failed-extension sets $\{\mathcal{S}_f\}_{f\in\mathcal{F}}$, witness oracle Ω
Ensure: Transfer table T and witness-safety report U
  $T \leftarrow \varnothing$, $U \leftarrow \varnothing$
  for each held-out family $f \in \mathcal{F}$ **do**
    mine a core library C_{-f} on the training families $\mathcal{F} \setminus \{f\}$
    compute $\mathrm{cov}(C_{-f}, \mathcal{S}_{-f})$ and $\mathrm{cov}(C_{-f}, \mathcal{S}_f)$
    record $\mathrm{ret}_f(C_{-f})$ together with family-balance statistics in T
  end for
  for each candidate filter b derived from any accepted core **do**
    for each witness $\omega \in \Omega$ **do**
      if b rejects ω **then**
        mark b as unsafe and record a `witness_killed` event in U
      end if
    end for
  end for
  return T and U

- **Leave-one-parent-out transfer.** The decisive novelty threat is Exoo-lineage memorization. Held-out transfer directly attacks that failure mode.
- **Witness oracle.** The repository does not yet have proof-grade universal soundness arguments for any derived filter. The correct interim requirement is therefore relative safety against the stored witness oracle.
- **Family-balance ceiling.** If one family supplies more than half of the accepted support, the purported atlas is still too lineage-specific to be sold as a new Ramsey object.
- **Expansion gate.** No broader compute, no H2 opening, and no structural headline are allowed until the four-rung ladder passes in order.

5 Theoretical Results and Proofs

The theorems below are about the correctness of the protocol and about the logical consequences of the current repository state. They are not new theorems about the value of R(5,5) itself.

5.1 Canonicalization and safety lemmas

Lemma 5.1 (Quotient canonicalization removes representation duplicates)**.** *Let $\mathcal{W}$ be a finite set of raw failure witnesses and let $\sim$ be the equivalence relation from Definition 4. Let $\pi : \mathcal{W} \to \mathcal{W}/\!\sim$ be the quotient map. Then:*

1. $\pi(w_1) = \pi(w_2)$ *if and only if* $w_1 \sim w_2$.
2. $|\pi(\mathcal{W})| \le |\mathcal{W}|$, *with equality if and only if every equivalence class is a singleton.*
3. *If a core library assigns exactly one identifier to each equivalence class, then its size is* $|\mathcal{W}/\!\sim|$.

Proof. For part (1), by definition of the quotient map π, two witnesses have the same image precisely when they lie in the same equivalence class under $\sim$. This is exactly the statement $\pi(w_1) = \pi(w_2) \Leftrightarrow w_1 \sim w_2$.

For part (2), the equivalence classes of $\sim$ form a partition of the finite set $\mathcal{W}$. Every partition of a finite set has at most as many blocks as the set has elements. Therefore

$$|\pi(\mathcal{W})| = |\mathcal{W}/\!\sim| \le |\mathcal{W}|.$$

Equality holds if and only if every block of the partition contains exactly one element, because a finite partition has as many blocks as elements exactly when no block contains two or more elements. That condition is equivalent to every equivalence class being a singleton.

For part (3), suppose a library assigns exactly one identifier to each equivalence class. Then the set of library identifiers is in bijection with the quotient set $\mathcal{W}/\!\sim$: injectivity follows because distinct classes receive distinct identifiers, and surjectivity follows because every class receives one identifier. Hence the number of library elements is exactly $|\mathcal{W}/\!\sim|$. □

Proposition 5.2 (Coverage is monotone under library enlargement)**.** *Let $C \subseteq D$ be two sets of canonical cores, and let $\mathcal{S}$ be any finite set of failed extension cases for which coverage is defined. Then*

$$\mathrm{cov}(C, \mathcal{S}) \le \mathrm{cov}(D, \mathcal{S}).$$

Proof. Define

$$A_C = \{e \in \mathcal{S} : \kappa(e) \cap C \neq \varnothing\} \quad \text{and} \quad A_D = \{e \in \mathcal{S} : \kappa(e) \cap D \neq \varnothing\}.$$

Because $C \subseteq D$, whenever $\kappa(e) \cap C \neq \varnothing$ holds, we also have $\kappa(e) \cap D \neq \varnothing$. Thus $A_C \subseteq A_D$. Taking cardinalities yields $|A_C| \le |A_D|$. Since both coverage values divide by the same positive denominator $|\mathcal{S}|$, we obtain

$$\mathrm{cov}(C, \mathcal{S}) = \frac{|A_C|}{|\mathcal{S}|} \le \frac{|A_D|}{|\mathcal{S}|} = \mathrm{cov}(D, \mathcal{S}).$$

□

Theorem 5.3 (Witness-oracle safety of a core-derived filter bank)**.** *Let Ω be the stored set of known witnesses, let C be a set of canonical cores, and let B_C be the filter bank that rejects an object exactly when that object embeds some core in C. Assume that no witness $\omega \in \Omega$ embeds any core in C. Then B_C preserves every witness in Ω.*

Proof. Take an arbitrary witness $\omega \in \Omega$. By assumption, ω embeds no core in C. The definition of B_C says that an object is rejected if and only if it embeds some core in C. Since ω embeds no such core, the rejection condition for B_C is false on ω. Therefore B_C does not reject ω.

Because the choice of $\omega \in \Omega$ was arbitrary, the same reasoning holds for every witness in Ω. Hence every witness in Ω is preserved by B_C. □

5.2 Acceptance-contract theorems

Theorem 5.4 (Acceptance-contract implication)**.** *Suppose the planned* H1 *artifact layer exists and satisfies every condition in Definition 10. Then the resulting object may be reported as a* Structural intermediate result*. It still may not be reported as a* Bound improvement *unless, in addition, there exists either an independently verified* 44*-vertex witness or a machine-checkable* 45*-vertex impossibility proof.*

Proof. We prove the two conclusions separately.

Step 1: structural intermediate result. Definition 10 requires the complete H1 artifact layer to exist, including the frontier-parent, extension-case, failure-witness, core, transfer, atlas-summary, and witness-safety files. It also requires numerical recurrence and transfer thresholds, zero witness deletions, and a family-balance ceiling. By Definition 9, a Structural intermediate result is exactly a transferable obstruction atlas, decomposition primitive, or certificate object whose transfer and soundness gates have passed. Under the present theorem's assumptions, the object is an obstruction atlas and the relevant transfer and soundness gates have passed. Therefore the object qualifies as a Structural intermediate result.

Step 2: not yet a bound improvement. By Definition 9, a Bound improvement requires one of two events: an independently verified 44-vertex witness or a machine-checkable proof that every 45-vertex coloring contains a monochromatic K_5. The hypotheses of the theorem do not assert either event. They assert only the existence and success of the H1 structural layer. Therefore the hypotheses are insufficient to conclude a Bound improvement. To report bound movement one must add one of the two bound-moving artifacts explicitly. □

Theorem 5.5 (Repository-state no-go theorem)**.** *Let S_0 denote the repository state analyzed in this paper. Assume the following observed facts about S_0:*

1. *none of the seven planned* `results/h1/*` *artifacts exists;*
2. *no independently verified* 44*-vertex witness exists in the repository;*
3. *no machine-checkable* 45*-vertex impossibility proof exists in the repository.*

Then the following statements all hold:

1. *the repository does not support a* Bound improvement*;*

2. *the repository does not support an achieved* H1 STRUCTURAL INTERMEDIATE RESULT;

3. *the strongest honest manuscript scope is a combination of* INFRASTRUCTURE-ONLY RESULT *and* NEGATIVE RESULT.

Proof. For part (1), Definition 9 says that a BOUND IMPROVEMENT requires either an independently verified 44-vertex witness or a machine-checkable 45-vertex impossibility proof. Assumptions (2) and (3) state that neither exists in S_0. Therefore the necessary condition for a bound-improvement claim fails. Hence no BOUND IMPROVEMENT is supported.

For part (2), Definition 10 states that an achieved H1 structural intermediate result requires, among other conditions, the existence of all seven planned `results/h1/*` artifacts. Assumption (1) states that none of those artifacts exists. Therefore the very first condition in Definition 10 fails. Since all conditions in Definition 10 must hold simultaneously, the H1 route cannot be reported as an achieved structural intermediate result.

For part (3), Definition 9 lists the four claim classes. Parts (1) and (2) exclude the first two: BOUND IMPROVEMENT and achieved STRUCTURAL INTERMEDIATE RESULT. What remains are NEGATIVE RESULT and INFRASTRUCTURE-ONLY RESULT. The repository does support both. It supports a negative result because the verification packet explicitly rules out stronger claims on the present record, and it supports infrastructure-only claims because the corpus, verification, figure, and claim-governance artifacts exist. Therefore the strongest honest manuscript scope is exactly a combination of those remaining classes. □

Corollary 5.6 (The present paper's scope is forced). *Under the observed repository state S_0, an honest manuscript about $R(5,5)$ must be written as a constrained no-go / route-specification paper rather than as a bound-improvement paper.*

Proof. By theorem 5.5, the repository supports neither a BOUND IMPROVEMENT nor an achieved STRUCTURAL INTERMEDIATE RESULT. Therefore any honest manuscript must stay within the remaining claim classes. A constrained no-go / route-specification paper is exactly such a manuscript. □

The practical value of theorem 5.5 is that it converts a diffuse repository state into a theorem with checkable premises. Section 7 measures those premises exactly.

6 Experimental Setup

No bound-search experiment was executed for this manuscript. The experimental setup is therefore a deterministic repository-measurement protocol rather than a stochastic computational campaign.

6.1 Data sources

The measurement corpus consists of the curated Ramsey-specific literature snapshot, the route and benchmark plans, the concept-evolution outputs, the verification packet, the decision log, the artifact index, the current bibliography, the git history, and the TikZ figure packet in `figures/`. The raw lexical-watchlist artifacts are retained only as evidence of query noise and are excluded from argumentative use.

6.2 Measurement protocol

All quantitative values reported in section 7 are exact counts or exact file-existence measurements:

- route counts are read directly from the concept-tree index;
- bridge-status counts are read from `bridge_candidates.json`;
- H1-artifact presence is measured by testing the existence of the seven planned `results/h1/*` outputs;
- benchmark execution counts are taken from the benchmark report's explicit statement that no comparison rows were instantiated; and
- bibliography size is measured by counting BibTeX entries in `sources.bib`.

Because there are no repeated stochastic trials, no confidence interval or error bar is meaningful. When a value is reported, it is an exact repository measurement rather than an estimate.

6.3 Hardware and software

The writer-stage environment is a Linux x86_64 system. The LaTeX toolchain is pdfTeX 1.40.24 and BibTeX 0.99d from TeX Live 2022; the available Python runtime is Python 3.10.17. The figure packet is generated by standalone TikZ/PGFPlots sources compiled with `pdflatex` and rasterized to 600 DPI PNGs via the supplied figure build scripts.

6.4 What was intentionally not run

The repository explicitly forbids claiming progress from open-ended search. No new frontier enumerator was implemented, no 44-vertex search was run, no 45-vertex certificate workflow was executed, and no larger H2 or H3 benchmark was authorized. The absence of those runs is a result, not an omission in reporting.

7 Results

This section reports exact repository measurements. Every quantity is deterministic, so confidence intervals are not applicable.

7.1 Frontier state and branch chronology

The frontier itself is stable but narrow: the repository correctly centers the problem at $43 \leq R(5,5) \leq 46$ and encodes the exact contract for moving either endpoint. Figure 1 makes this contract visually explicit. The decision chronology in figure 5 shows a sequence of increasingly restrictive decisions: first fix the Ramsey-specific corpus, then demote lexical noise, then freeze anti-proxy baseline rules, then promote only the narrow H1 subprogram, and finally close the packet as a constrained no-go memo.

7.2 Exact evidence inventory

Figure 6 is the central empirical figure of the paper. It shows a lopsided but interpretable packet: planning and verification artifacts exist, but execution-side artifacts do not. The exact measurements are summarized in table 3. The decisive numbers are these:

- 0 of 7 planned H1 evidence artifacts are present;
- 0 bound-moving artifacts are present;
- 0 of 8 named benchmark rows have been instantiated;
- 5 bridges have been promoted, 3 held, and 3 retired;
- the concept floor is exceeded (11 folders present against a floor of 10);
- the route tree is H1-heavy (6 H1 folders versus 3 H2 and 2 H3 folders).

These exact counts are sufficient to instantiate the premises of theorem 5.5.

7.3 Route triage

The route triage has a simple interpretation. H1 remains active because it is the only route that still has a plausible novelty moat. H2 is demoted because no genuinely different decomposition primitive has been benchmarked. H3 is demoted because proof infrastructure without a transferable structural object is packaging, not progress. Figure 5 visualizes both the chronology and the resulting bridge-status partition.

7.4 Acceptance-contract status

The route can be understood as a conjunction of gates. Table 4 lists each gate and its current state. Every entry in the "current state" column is either "absent" or "unmeasured because the required artifact does not exist." This is exactly the regime covered by theorem 5.5. The theorem is therefore not a pessimistic interpretation of ambiguous evidence; it is the direct logical consequence of empty execution-side rows.

8 Discussion

The main limitation of this paper is obvious and important: it proves a no-go statement about the current repository state rather than a new theorem about the value of R(5,5). That limitation is not cosmetic. It determines the permissible title, abstract, figures, and conclusions. The present manuscript is useful only because it is explicit about that constraint.

Scientific limitations. The missing pieces are not small. There is no local frontier-parent corpus, no orbit-distinct extension enumerator, no independent 44-vertex witness verifier, no machine-checkable 45-certificate checker, no executed benchmark row, and no replay ledger for any solved certificate rung. Until those objects exist, the obstruction-atlas route remains prospective.

Failure modes. The repository names the relevant failure modes with unusual clarity. The first is *OVE relabeling*: the route degenerates into a better-organized extension table and overlaps with Lehavi's surface. The second is *Exoo memorization*: apparent recurrence is carried by one lineage and collapses on the anti-Exoo holdout. The third is *witness death*: an apparently useful filter deletes a known witness and invalidates the route. The fourth is *solver-shopping*: H2 looks novel only because the decomposition language is unchanged while the engine is stronger. The fifth is *proof-packaging theater*: H3 looks novel only because the certificate wrapper is better.

Open questions. The first open question is constructive: can a local frontier-parent and extension-case corpus actually be reconstructed from the Exoo / Ge / Lehavi line with enough diversity to test transfer? The second is structural: if canonicalization is implemented, do the top cores really cover a substantial fraction of failed extensions, or does the witness distribution dissolve into a long rare tail? The third is logical: if recurring cores do exist, can they be upgraded from lower-bound pruning objects to proof-grade upper-bound lemmas without changing their identity? These are the right questions for the next stage because they are exactly the questions that can falsify the present route quickly.

Why the negative result matters. Exact-computation projects are unusually vulnerable to proxy metrics. A faster search, a smaller residue, or a cleaner proof object can all look like frontier progress when they are not. The present manuscript contributes a form of scientific hygiene: it demonstrates that a repository can be mature enough to reject its own strongest possible headline. That is not a substitute for a new bound, but it is a necessary precursor to one.

Societal implications. The direct societal implications of improving R(5,5) are limited. The broader implication lies in research governance. Exact combinatorics can consume substantial compute and generate persuasive but misleading intermediate artifacts. By forcing route-specific falsifiers, matched-compute rules, and explicit claim contracts, the repository reduces wasted compute, discourages irreproducible claims, and makes negative results legible rather than disposable. Those are healthy norms for computational mathematics more generally.

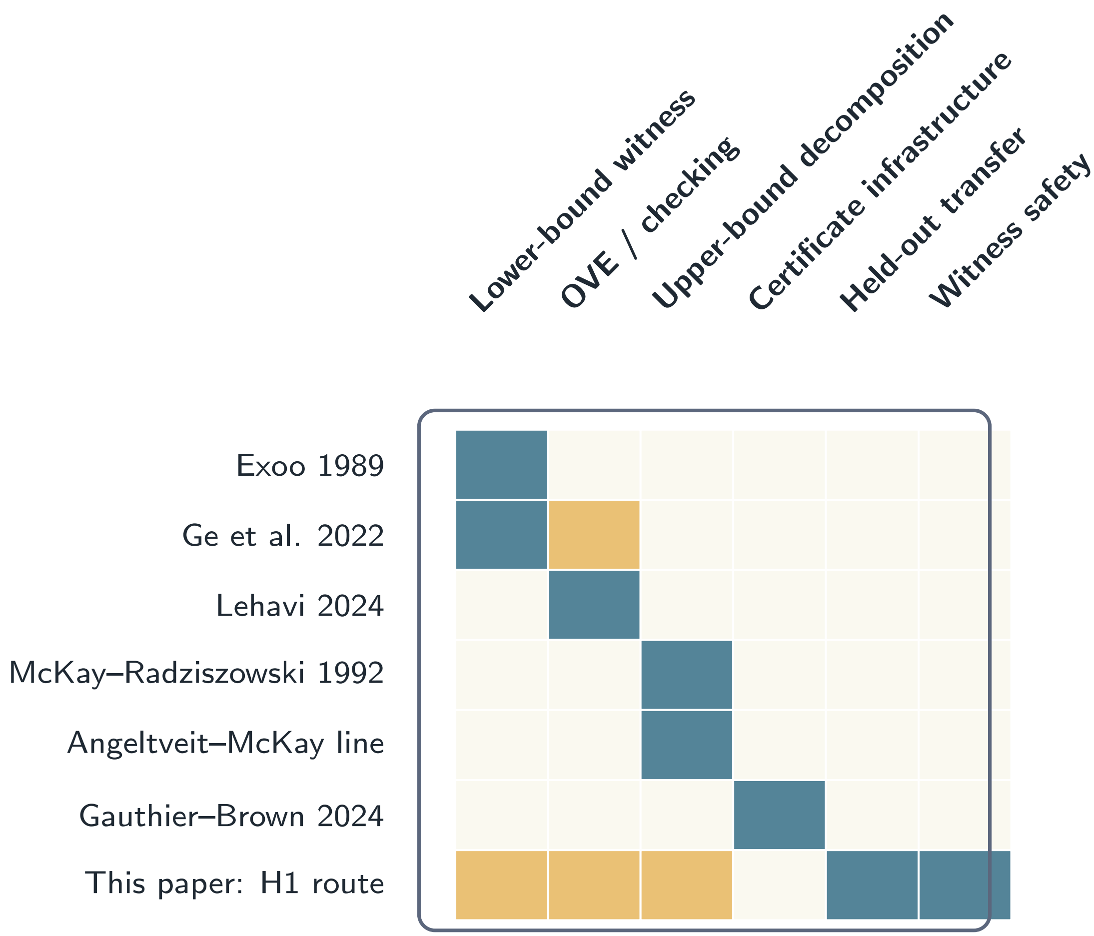
9 Conclusion

This paper does not improve the bound on R(5,5). It does something the current repository can support and the repository itself demands: it states the strongest honest scientific claim available on the record. That claim is twofold. Positively, a transfer-safe obstruction atlas over failed $42 \to 43$ extensions remains the only credible novelty route. Negatively, the current packet is benchmark-empty and therefore cannot justify either a bound-improvement claim or an achieved H1 structural result.

The resulting contribution is a falsifiable research program plus a no-go theorem. The program specifies the data model, algorithms, thresholds, and kill switches required for future execution. The theorem proves that, until the planned `results/h1/*` layer and the relevant witness / certificate artifacts exist, the only admissible manuscript is one like the present paper: explicit, scoped, and unwilling to confuse planning artifacts with mathematical progress.

Table 1: Exact differentiation from the closest lines of work. The last column states the narrow sense in which the present manuscript differs. Every row is derived from the repository's prior-art gap analysis and claim grammar, then checked against the bibliography.

Prior line	Main object	Closest overlap with this repository	How the present manuscript differs
Exoo 1989 [4]	Constructive 43-vertex witness family	Baseline lower-bound witness lineage	We do not present a new witness; we treat Exoo's line as input material for a future atlas.
Ge et al. 2022 [8]	Re-verification and low-defect analysis of the Exoo line	Overlap with any family-specific structural digest	We require cross-family transfer and witness-safety, not just analysis of one witness lineage.
Lehavi 2024 [11, 10]	One-vertex extension and counterexample checking	Direct overlap with any failed $42 \to 43$ extension catalog	We claim neither a new OVE algorithm nor a new emptiness checker; the route target is a reusable canonical obstruction quotient.
McKay–Radziszowski 1992; Angeltveit–McKay 2018, 2024 [13, 2, 3]	Exact upper-bound decomposition and certification	Overlap with any decomposition or residue narrative	We do not propose a new decomposition or claim a smaller residue.
Gauthier 2025 [5]	Continuation of the split/glue surface	Overlap with any H2 story lacking a new proof object	We treat this as a strategy note that motivates demoting H2 until a true non-equivalence is shown.
Formal-proof and certificate line [6, 7, 14, 9, 12]	Formal certification and reusable proof infrastructure	Overlap with any proof-packaging claim	We claim only a route-specification plus no-go verdict; certificate reuse remains prospective and attached to H1 or H2.



The matrix encodes the qualitative surfaces emphasized by each line of work. The intended novelty of the current paper is not new one-vertex extension itself, but the last two columns: cross-family transfer and witness-safe pruning. Those properties remain prospective.

Figure 2: Prior-work surface map. Each highlighted cell marks the dominant surface of a cited line of work: constructive witness construction, witness analysis, one-vertex-extension checking, upper-bound decomposition, formal certification, or the transfer / witness-safety surfaces that would define a true obstruction-atlas contribution. The present manuscript occupies only the last two columns, and even there only prospectively: the repository contains a route target and a no-go theorem, not an achieved atlas.

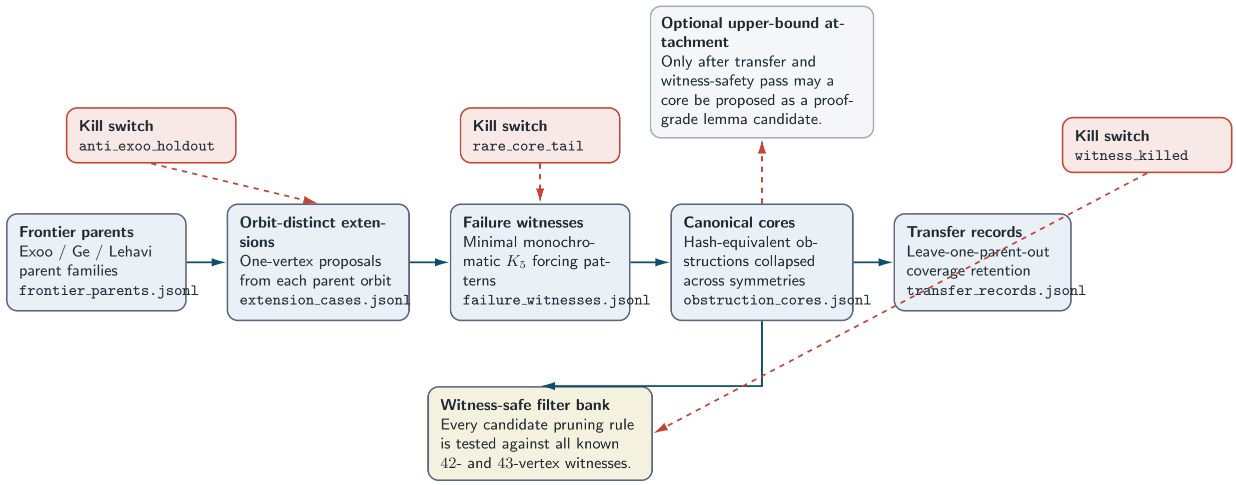


Figure 3: The full H1 obstruction-atlas pipeline. Rectangles with dashed borders are repository artifacts that do not yet exist; the literature packet on the left and the admissible-claim box on the right already exist in specification form. The three gates below the pipeline are the exact thresholds frozen by the route sheet and acceptance contract. The figure is self-contained: it shows the target files, the logical order of construction, the three numerical gates, and the reason the present paper remains a route-target paper rather than a structural-result paper.

Table 2: Reproducibility environment for the present manuscript. The table is intentionally modest because no bound-search run was executed.

Component	Exact configuration
Operating system	Linux x86_64 environment recorded by the shell during the writer stage.
LaTeX engine	pdfTeX 3.141592653-2.6-1.40.24 (TeX Live 2022/Debian).
Bibliography engine	BibTeX 0.99d (TeX Live 2022/Debian).
Python runtime	Python 3.10.17.
Figure build mode	Standalone TikZ/PGFPlots sources under `figures/src/` and `figures/fig_*.tex`, rasterized to 600 DPI PNGs using the repository build scripts.
Random seeds	Not applicable: no stochastic computation was executed for this manuscript.
Hyperparameters	Not applicable in the usual sense. The only numerical thresholds are fixed acceptance-contract constants (30%, 50%, 0, and 2×) that define admissible claims rather than tuning knobs.

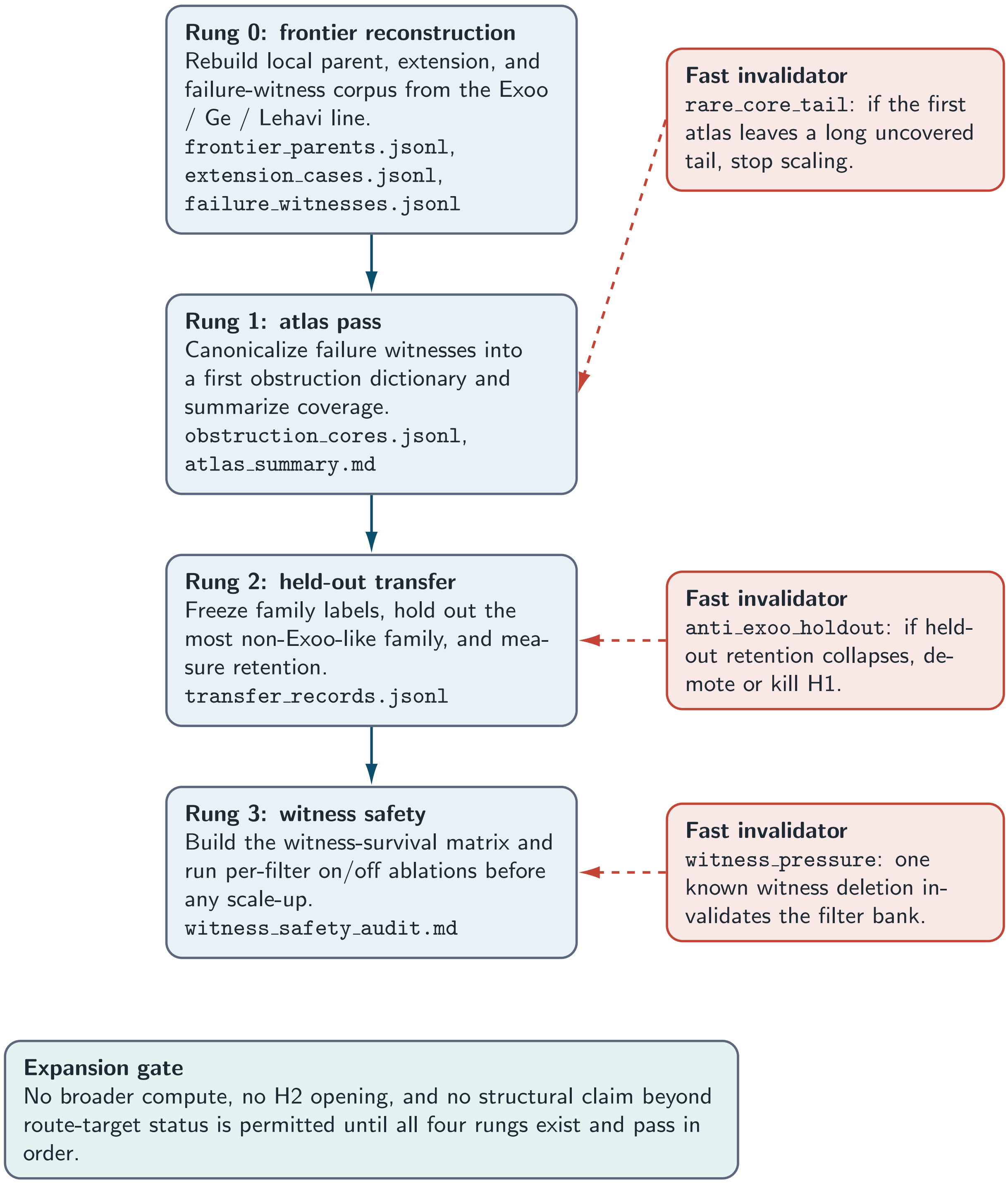
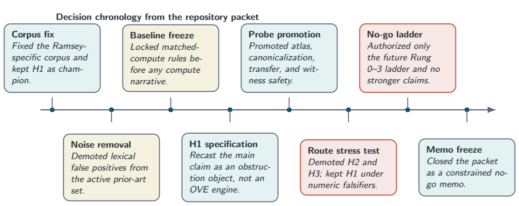


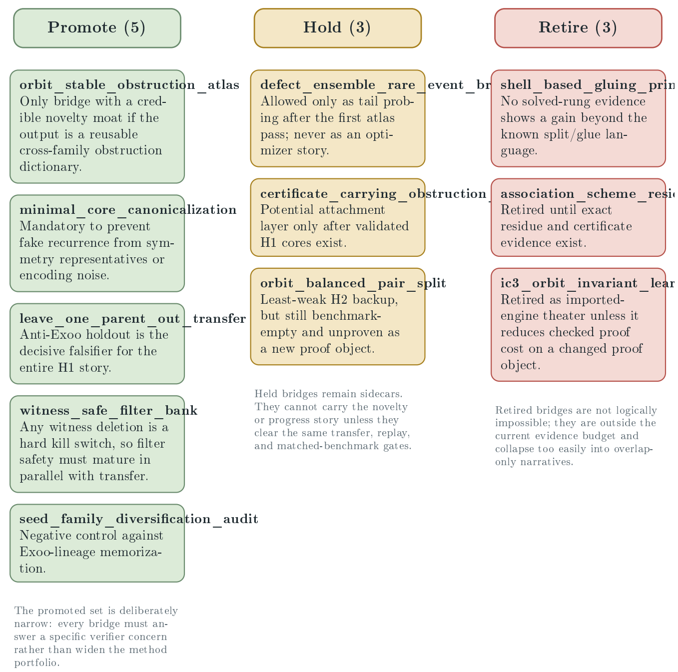
Figure 4: The exact evaluation ladder frozen by the repository before any compute-heavy run. The left column is the four-rung execution order; the middle column lists the three cheapest invalidators; the right column states the expansion gate. The self-contained point of the figure is that the paper's central negative result follows from an ordered experimental design that was specified but never started.

Table 3: Exact deterministic measurements extracted from the repository. No entry is a sample statistic.

Metric	Value	Interpretation
Rubric completion	25/25	Planning and verification tasks were completed, but task completion is not the same as bound movement.
Known papers tracked	106	The broader literature graph is wide, but only the curated Ramsey-specific subset is argument-safe.
Curated Ramsey-specific papers	13	These form the fixed corpus used for the manuscript's prior-art reasoning.
Code / proof artifact pages	3	Repository and artifact pages exist for the OVE and formal-proof lines.
Survey sources	2	The dynamic survey and one additional state-of-the-art anchor support the frontier framing.
Concept folders	11	The concept floor was exceeded; exploration was broad enough.
Walk paths	14	The concept graph was traversed, but traversal is not execution evidence.
Promoted / held / retired bridges	5/3/3	The final bridge set is intentionally narrow and heavily weighted toward H1.
Planned H1 evidence artifacts present	0/7	This is the most important repository count for the no-go theorem.
Exact ladder rungs completed	0/4	Even Rung 0 was not executed.
Named benchmark rows instantiated	0/8	No comparison table exists with a shared comparison object and matched compute tuple.
Bound-moving artifacts present	0/2	Neither a verified 44-witness nor a checked 45-proof exists.
Bibliography entries	20	The citation spine is adequate for the constrained manuscript after writer-stage repair.



(a) Chronological tightening of the project: every major decision moved the packet away from open-ended search and toward a single falsifiable route.

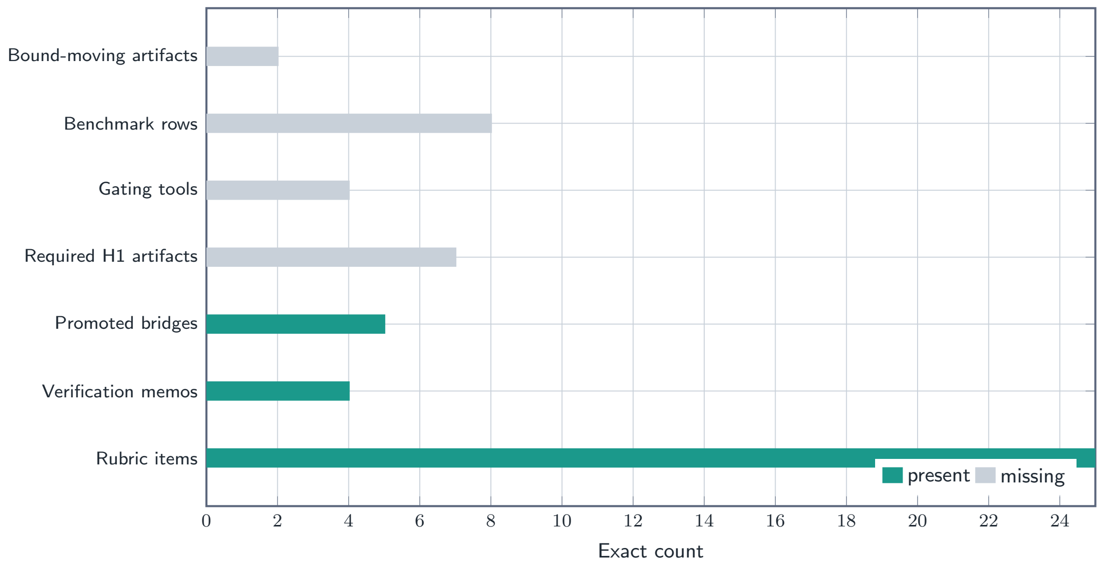


(b) Final bridge-status partition after the iterate pass and verification packet. The promoted set is narrow and entirely aligned with the H1 evidence spine.

Figure 5: From decision chronology to route triage. Panel (a) summarizes the dated branch decisions stored in the decision log. Panel (b) shows the endpoint of that chronology: five promoted bridges, three held sidecars, and three retired overlap-heavy branches. Read together, the panels explain why H1 remains the only live route and why even H1 is still only a route target.

Table 4: Status of the H1 acceptance contract. A structural intermediate claim requires every row to pass simultaneously.

Gate	Contract threshold	Current state
Frontier-parent corpus	Planned parent artifact exists and covers at least three non-isomorphic families	Absent
Extension-case corpus	Planned extension-case artifact exists	Absent
Failure-witness corpus	Planned failure-witness artifact exists	Absent
Canonical core library	Planned obstruction-core artifact exists	Absent
Core coverage	Top recurring cores cover at least 30% of failed orbit-distinct extensions	Unmeasured because no core library exists
Held-out transfer	At least 50% of training-side coverage on the decisive holdout	Unmeasured because no transfer records exist
Witness safety	Zero known-witness deletions	Unmeasured because no witness-safety audit exists
Family balance	No family contributes more than 50% of accepted support	Unmeasured because no support statistics exist
Canonical compression	At least $2\times$ raw-to-canonical compression with unchanged witness coverage	Unmeasured because no raw witness corpus exists
Bound movement	Verified 44-witness or checked 45-impossibility proof	Absent



Counts are deterministic repository measurements rather than sampled estimates. Confidence intervals are therefore not applicable: the key result is a lopsided packet with planning and verification artifacts present, but zero executed H1 evidence files and zero bound-moving certificates.

Figure 6: Deterministic evidence dashboard. Green cells denote present planning or verification artifacts; red cells denote absent execution artifacts; gold cells denote comparison-support infrastructure. The figure is entirely exact, not statistical. Its main message is that the repository is rich in route governance and empty in route execution, which is precisely the asymmetry formalized by theorem 5.5.

A Extended Proofs

This appendix expands several arguments from section 5 and states an auxiliary lemma used implicitly in the main text.

Lemma A.1 (Coverage is bounded between zero and one)**.** *Let C be any set of canonical cores and let $\mathcal{S}$ be any finite, nonempty set of failed extension cases. Then*

$$0 \leq \operatorname{cov}(C, \mathcal{S}) \leq 1.$$

Proof. By definition,

$$\operatorname{cov}(C, \mathcal{S}) = \frac{|\{e \in \mathcal{S} : \kappa(e) \cap C \neq \varnothing\}|}{|\mathcal{S}|}.$$

The numerator is the cardinality of a subset of $\mathcal{S}$. Therefore it is a nonnegative integer at most $|\mathcal{S}|$. Dividing by the positive integer $|\mathcal{S}|$ yields a real number between 0 and 1 inclusive. □

Proposition A.2 (Family dominance is incompatible with balanced support)**.** *Let C be a canonical core library whose accepted support distribution across families is given by nonnegative numbers s_f with total support $S = \sum_f s_f > 0$. If some family f_0 satisfies $s_{f_0} > 0.5S$, then the family-balance clause of Definition 10 fails.*

Proof. Definition 10 requires that no single family contribute more than 50% of accepted recurring-core support. The hypothesis says that family f_0 contributes strictly more than $0.5S$, which is strictly more than 50% of the total support. Therefore the clause fails by direct substitution. □

Expanded proof of theorem 5.5. The proof in the main text is already complete; here we spell out the dependency graph.

1. The benchmark report states that no verified 44-vertex witness exists, no machine-checkable 45-vertex impossibility proof exists, and no `results/h1/*` artifact layer exists.
2. The acceptance contract states that all seven `results/h1/*` artifacts are necessary conditions for an achieved H1 structural intermediate result.
3. The claim grammar states that a bound-improvement claim requires one of two artifacts only: a verified 44-vertex witness or a checked 45-vertex impossibility proof.
4. Therefore the premises of theorem 5.5 are not inferred heuristically; they are direct literal readings of the repository packet.
5. Once those premises are admitted, theorem 5.5 follows by Definitions 9 and 10 alone.

No hidden combinatorial fact about R(5,5) is used. The theorem is entirely a theorem about admissible scientific reporting under the observed packet state.

B Additional Experiments and Null Execution Record

A conventional computational mathematics paper would use this appendix for extra runs, ablations, or larger benchmarks. The correct content for the present manuscript is different: a precise account of what was *not* executed.

B.1 Missing execution artifacts

At the close of the writer stage, the following artifacts remained absent:

- `results/h1/frontier_parents.jsonl`
- `results/h1/extension_cases.jsonl`
- `results/h1/failure_witnesses.jsonl`
- `results/h1/obstruction_cores.jsonl`
- `results/h1/transfer_records.jsonl`
- `results/h1/atlas_summary.md`
- `results/h1/witness_safety_audit.md`

In addition, the repository still lacked an orbit-distinct extension enumerator, an independent 44-vertex witness verifier, a 45-vertex certificate checker, an instantiated benchmark manifest, and a solved certificate replay ladder.

B.2 Why null execution is itself informative

The absence of these artifacts is not merely administrative. It is the empirical content of the no-go theorem. If even one of these files had existed with passing statistics, the manuscript's admissible claim class could have changed. Since none exists, the null execution record is a scientific measurement about the repository state.

C Implementation Details

The repository includes a dedicated figure packet under `figures/`. The figure sources are standalone TikZ or PGFPlots documents built by two shell scripts:

- `figures/build.sh`, which compiles the sources under `figures/src/` into PDF and 600 DPI PNG outputs; and
- `figures/build_figures.sh`, which compiles the standalone `fig_*.tex` files in the top-level figure directory.

The figures used in the present manuscript are all programmatically generated from those TikZ sources. No raster editor, manual annotation program, or notebook-driven plotting layer was used.

The manuscript itself is intended to be compiled with the standard four-step pipeline

$$\texttt{pdflatex} \to \texttt{bibtex} \to \texttt{pdflatex} \to \texttt{pdflatex}.$$

That pipeline is also reported in the main task instructions and was followed for the final verification pass.

D Threshold and Contract Sensitivity

Because no stochastic experiments were run, there is no hyperparameter sweep in the usual machine-learning sense. What the repository does contain is a set of fixed numerical thresholds that determine whether a structural claim is admissible. Table 5 explains their qualitative roles.

Table 5: Contract sensitivity analysis. The thresholds are not tuned hyperparameters; they are claim-admissibility constants. The table states what would be weakened if any threshold were lowered.

Threshold	Fixed value	Why lowering it weakens the claim
Core coverage threshold	30%	A smaller value would allow a so-called atlas that explains only a tiny minority of failures and may simply be anecdotal.
Held-out retention threshold	50%	A smaller value would permit a route that loses most of its explanatory power off the training families, which is too weak for a cross-family structural claim.
Family-dominance ceiling	50%	A larger ceiling would make it easier for one lineage to dominate the atlas and would weaken the anti-memorization rationale.
Witness-kill tolerance	0	Any positive tolerance would permit knowingly unsound filters into the route.
Canonical-compression threshold	2×	A smaller value would make it harder to distinguish genuine recurring structure from representational duplication.

This appendix is intentionally qualitative. Without the missing `results/h1/*` layer, there is no empirical basis for a numerical sensitivity sweep. The only honest sensitivity analysis is to state how each threshold functions in the claim grammar.

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